

TA KWU LING, Hong Kong

WMO: 450320

Lat: 22.5291N

Lon: 114.1560E

Elev: 47

StdP: 14.67

Time Zone: 8.00 (E08)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	42.2	44.8	22.7	17.1	55.5	29.0	23.0	53.5	14.5	52.1	13.5	52.7	4.3	350	0.306

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	93.6	80.1	92.4	79.9	91.2	79.8	83.1	89.0	81.9	88.3	81.1	87.3	5.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.8	165.5	86.0	80.2	157.1	84.8	79.5	153.5	84.1	47.3	89.1	45.9	88.7	45.0	88.0	95.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.5	11.7	10.0	DB	36.8	97.0	3.5	1.6	34.3	98.1	32.2	99.0	30.3	99.9	27.8	101.0
			WB	34.1	87.7	2.8	5.0	32.0	91.4	30.4	94.3	28.8	97.2	26.7	100.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	73.7	59.5	62.7	66.6	73.5	79.3	82.6	83.9	83.6	82.0	77.4	70.4	62.1
	DBStd	10.04	6.56	7.58	6.06	5.11	3.78	2.69	2.18	2.33	3.43	3.69	5.84	6.38
	HDD50	16	8	6	0	0	0	0	0	0	0	0	0	2
	HDD65	524	191	118	55	6	0	0	0	0	2	0	20	132
	CDD50	8667	304	361	513	706	910	978	1051	1042	961	850	614	377
	CDD65	3697	21	53	103	262	445	528	586	577	512	385	183	42
	CDH74	34379	44	143	314	1439	3900	5810	6929	6571	5331	2893	886	119
	CDH80	12095	1	7	27	326	1223	2145	2887	2635	1923	774	141	6
Wind	WSAvg	4.7	4.9	5.3	5.4	5.5	5.0	4.2	4.1	3.8	4.5	4.6	4.8	4.9
Precipitation	PrecAvg	72.3	1.1	1.7	2.7	6.1	9.6	11.5	12.1	13.8	9.5	2.1	1.5	1.0
	PrecMax	104.8	3.8	8.0	14.6	18.1	21.7	29.1	21.9	26.2	20.8	17.3	5.1	6.7
	PrecMin	35.9	0.0	0.0	0.1	0.2	1.0	2.8	3.8	1.6	1.1	0.0	0.0	0.0
	PrecStd	14.8	1.1	2.0	3.0	4.4	5.6	5.8	5.9	6.2	5.4	3.3	1.6	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.3	81.1	82.7	88.2	93.1	93.4	95.8	95.5	94.2	90.8	86.9	80.6
		MCWB	66.4	69.9	71.8	75.8	79.1	79.9	80.7	80.7	79.7	76.9	73.4	69.4
	2%	DB	74.5	78.3	80.2	85.9	90.5	91.9	93.4	93.5	92.1	88.1	83.7	77.3
		MCWB	64.6	68.7	71.4	75.0	78.3	79.8	80.1	80.6	79.2	75.3	71.7	66.2
	5%	DB	71.5	75.3	77.6	83.9	88.5	90.6	92.0	92.1	90.4	86.2	81.2	74.5
		MCWB	62.2	67.0	69.1	74.5	78.0	79.8	80.1	80.5	78.4	73.6	70.6	64.1
	10%	DB	68.5	72.5	75.0	81.7	86.3	88.9	90.5	90.5	88.7	84.4	79.0	71.6
		MCWB	60.9	66.0	68.0	73.9	77.4	79.6	80.1	80.5	77.8	72.7	69.4	63.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.9	72.8	74.3	78.2	81.0	84.2	85.2	90.0	82.8	80.1	76.3	72.6
		MCDB	74.8	78.1	80.1	84.1	87.6	88.8	88.2	90.6	88.2	85.0	82.6	76.8
	2%	WB	66.6	70.9	72.8	77.1	80.2	81.8	82.6	83.5	81.0	78.3	74.9	69.5
		MCDB	71.8	75.7	78.1	82.6	86.4	88.0	89.0	89.1	87.7	83.9	80.1	73.9
	5%	WB	64.6	69.1	71.4	76.0	79.5	81.0	81.6	81.9	80.2	77.1	73.7	67.2
		MCDB	69.2	73.8	75.9	81.5	85.4	86.9	88.1	88.3	86.5	82.6	78.4	72.1
	10%	WB	62.8	66.9	69.9	74.8	78.7	80.5	80.9	81.0	79.4	76.0	72.2	64.8
		MCDB	67.1	71.1	73.3	80.2	84.3	86.1	87.4	87.1	85.1	81.3	76.8	70.1
Mean Daily Temperature Range	5% DB	MDBR	14.1	12.3	11.6	11.2	11.1	10.9	12.3	12.4	12.4	13.4	14.1	15.2
		MCDBR	17.5	16.6	15.5	14.9	14.6	13.8	15.4	15.7	15.2	15.5	16.0	18.1
		MCWBR	7.8	8.0	7.4	5.8	4.8	4.4	4.6	5.5	4.7	5.1	5.9	7.8
	5% WB	MCDDBR	13.4	13.2	12.6	12.6	12.3	11.6	13.0	13.5	13.2	12.3	12.3	13.4
		MCWBR	7.5	7.2	6.9	5.9	4.8	4.6	5.2	6.4	4.9	5.2	5.8	7.5
Clear-Sky Solar Irradiance	taub	0.584	0.638	0.769	0.747	0.587	0.555	0.542	0.608	0.641	0.682	0.577	0.568	
	taud	1.792	1.718	1.508	1.587	1.959	2.058	2.089	1.893	1.798	1.664	1.871	1.851	
	Ebn at Noon	214	213	193	201	235	241	244	229	218	199	215	213	
	Edh at Noon	63	72	93	87	60	54	52	63	69	75	58	58	
All-Sky Solar Radiation	RadAvg	981	989	1056	1159	1395	1433	1679	1580	1489	1391	1134	1027	
	RadStd	159	155	144	183	193	217	178	144	106	141	129	151	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	N/A	N/A	N/A	+0.45	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page